Problem Set 2 – Solutions

AI 501: Mathematics for Artificial Intelligence

Fall 2026

Solution 1.

(a)
$$x^T y = 2(-1) + (-1)(4) + 3(2) = 0.$$

(b)
$$\|x\| = \sqrt{4+1+9} = \sqrt{14}, \qquad \|y\| = \sqrt{1+16+4} = \sqrt{21}.$$

(c)
$$\|x - y\| = \left\| \begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \right\| = \sqrt{9+25+1} = \sqrt{35}.$$

(As a check, since $x \perp y$: $\|x-y\|^2 = \|x\|^2 + \|y\|^2 = 14 + 21 = 35$.)

(d)
$$\cos\theta = \frac{x^T y}{\|x\|\|y\|} = 0, \qquad \text{so} \qquad \boxed{\theta = \frac{\pi}{2}}.$$

Prediction. Since $x^T y = 0$ (neither positive nor negative), we expect a **right angle** before computing anything: x and y are orthogonal. Part (d) confirms this.

Solution 2. Let
$$w = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Orthogonality to u and v gives
$$u^T w = a + 2b - c = 0, \qquad v^T w = 2a - b = 0.$$

Thus $b = 2a$ and $c = a + 2b = 5a$. Taking $a = 1$,
$$\boxed{w = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}}.$$

Check: $u^T w = 1 + 4 - 5 = 0$ and $v^T w = 2 - 2 + 0 = 0$.

The answer is not unique: every nonzero scalar multiple of w also works, since a was a free choice. (Up to scaling, however, the direction of w is unique.)

Solution 3.

(a) **Sometimes.** Orthogonality says nothing about lengths. Orthogonal vectors can have equal lengths, e.g. $x = (1, 0)$, $y = (0, 1)$, but need not:

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

are orthogonal with $\|x\| = 1 \neq 2 = \|y\|$.

(b) **Sometimes.** Orthogonal vectors can have equal lengths, e.g. $x = (1, 0)$, $y = (0, 1)$. But equal lengths do not imply orthogonality: $x = y = (1, 0)$ have $\|x\| = \|y\| = 1$ but $x^T y = 1 \neq 0$.

(c) **Always.**

$$x^T y = \|x\|\|y\| \cos\theta > 0 \quad \Rightarrow \quad \cos\theta > 0,$$

so θ is acute.

(d) **Always.** Similarly,

$$x^T y < 0 \quad \Rightarrow \quad \cos\theta < 0,$$

so θ is obtuse.

(e) **Always.**

$$\|x + y\|^2 = \|x\|^2 + 2x^T y + \|y\|^2 = \|x\|^2 + \|y\|^2.$$

Solution 4.

For orthogonal u, v,

$$\|\alpha u + \beta v\|^2 = \alpha^2 \|u\|^2 + \beta^2 \|v\|^2.$$

Since $\|u\| = 3$ and $\|v\| = 4$:

(a)
$$\|u + v\| = \sqrt{9 + 16} = \boxed{5}$$

(b)
$$\|2u - v\| = \sqrt{4(9) + 16} = \sqrt{52} = \boxed{2\sqrt{13}}$$

(c)
$$\|3u + 4v\| = \sqrt{9(9) + 16(16)} = \sqrt{81 + 256} = \boxed{\sqrt{337}}$$

Solution 5.

(a) Both vectors have norm 5, so

$$\hat{x} = \frac{x}{\|x\|} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}, \qquad \hat{y} = \frac{y}{\|y\|} = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}.$$

(b)
$$\hat{x}^T \hat{y} = -\tfrac{12}{25} + \tfrac{12}{25} = 0.$$

(c)
$$x^T y = 3(-4) + 4(3) = 0,$$

so here both inner products equal 0. In general, $\hat{x}^T \hat{y} = \dfrac{x^T y}{\|x\|\|y\|}$: normalization rescales the inner product by the positive factor $1/(\|x\|\|y\|)$, so its value may change but its sign (positive, zero, or negative) does not.

(d) Normalization changes magnitude but preserves direction, and therefore preserves the angle between x and y (in particular, orthogonality): $\hat{x}^T \hat{y} = \cos\theta$.

Solution 6.

Since $y = 2x$, the vectors x and y point in the same direction. Since $x^T z = 0$ (computed below), z is orthogonal to x, and hence also to $y = 2x$.

(a) The pair (x, y) has angle 0.

(b) No pair has an angle strictly between 0° and 90°. (Under the rule $x^T y > 0 \Leftrightarrow$ acute from Problem 3(c), the pair (x, y) technically qualifies, since $x^T y = 28 > 0$; but its angle is exactly 0.)

(c) No pair has an obtuse angle: no inner product below is negative.

Verification.

$$x^T y = 2 + 8 + 18 = 28, \qquad \|x\| = \sqrt{14}, \qquad \|y\| = \sqrt{56} = 2\sqrt{14},$$

$$\cos\angle(x, y) = \frac{28}{\sqrt{14} \cdot 2\sqrt{14}} = 1,$$

so $\boxed{\angle(x, y) = 0}$. Also,

$$x^T z = 1(-2) + 2(1) + 3(0) = 0, \qquad y^T z = 2(-2) + 4(1) + 6(0) = 0,$$

so

$$\boxed{\angle(x, z) = \angle(y, z) = 90^\circ}.$$

Solution 7.

Student A is **incorrect**. Same direction does not imply zero distance. For example,

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

have distance 4.

Student B is also **incorrect**. For orthogonal vectors, the distance depends on their lengths. For example,

$$\left\| \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\| = \sqrt{2},$$

whereas

$$\left\| \begin{pmatrix} 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 4 \end{pmatrix} \right\| = 5.$$

Solution 8.

Recall

$$x_c = x - \mu \mathbf{1}, \qquad \mu = \frac{1}{n} \mathbf{1}^T x.$$

(a) The ith entry of x_c is $x_i - \mu$, so

$$\sum_{i=1}^{n} (x_i - \mu) = \sum_{i=1}^{n} x_i - n\mu = n\mu - n\mu = 0.$$

(b) The same computation in vector notation, using $\mathbf{1}^T \mathbf{1} = n$:

$$\mathbf{1}^T x_c = \mathbf{1}^T x - \mu\, \mathbf{1}^T \mathbf{1} = \mathbf{1}^T x - \frac{1}{n} (\mathbf{1}^T x)\, n = 0.$$

(Parts (a) and (b) say the same thing, since $\mathbf{1}^T x_c$ is the sum of the entries of x_c.)

(c) $\mathbf{1}^T x_c = 0$ means

$$\boxed{x_c \perp \mathbf{1}}.$$

Geometrically, $x = \mu \mathbf{1} + x_c$ splits x into a part along $\mathbf{1}$ and a part orthogonal to $\mathbf{1}$; centering removes the component of x in the $\mathbf{1}$ direction.

Solution 9.

Throughout, the standard deviation is $\operatorname{std}(x) = \|x - \mu \mathbf{1}\|/\sqrt{n}$, the standardized vector is $(x - \mu \mathbf{1})/\operatorname{std}(x)$, and the correlation coefficient is $\rho = \frac{1}{n} z_x^T z_y$ for standardized vectors z_x, z_y.

(a) Means:

$$\mu_x = \frac{1+2+3+4}{4} = 2.5, \qquad \mu_y = \frac{10+20+30+40}{4} = 25.$$

The centered vectors are $x - \mu_x \mathbf{1} = (-1.5, -0.5, 0.5, 1.5)$ and $y - \mu_y \mathbf{1} = (-15, -5, 5, 15)$, so

$$\operatorname{std}(x) = \sqrt{\frac{2.25 + 0.25 + 0.25 + 2.25}{4}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2} \approx 1.118,$$

$$\operatorname{std}(y) = 10 \operatorname{std}(x) = 5\sqrt{5} \approx 11.18.$$

(b) Since $y = 10x$, the two vectors have identical standardized versions:

$$z_x = z_y = \frac{2}{\sqrt{5}} \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix} = \begin{pmatrix} -3/\sqrt{5} \\ -1/\sqrt{5} \\ 1/\sqrt{5} \\ 3/\sqrt{5} \end{pmatrix}.$$

(c)
$$\operatorname{corr}(x, y) = \frac{1}{4}\, z_x^T z_y = \frac{1}{4} \cdot \frac{9+1+1+9}{5} = \boxed{1}.$$

Prediction for z. As the entries of x increase, the entries of z decrease, so we predict a *negative* correlation. In fact $z = 50\mathbf{1} - y = 50\mathbf{1} - 10x$, so z is a constant plus a negative multiple of x. Then $\mu_z = 25$, $z - \mu_z \mathbf{1} = (15, 5, -5, -15) = -(y - \mu_y \mathbf{1})$, and hence $z_z = -z_x$. Verifying,

$$\operatorname{corr}(x, z) = \frac{1}{4}\, z_x^T z_z = -\frac{1}{4} \cdot \frac{20}{5} = \boxed{-1}.$$

Solution 10.

(a)
$$v_3 = v_1 + v_2.$$

(b) No. Any combination using v_3 can be rewritten using v_1, v_2 alone, so

$$\operatorname{span}\{v_1, v_2, v_3\} = \operatorname{span}\{v_1, v_2\}.$$

(c)
$$\operatorname{span}\{v_1, v_2, v_3\} = \left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} : a, b \in \mathbb{R} \right\},$$

the plane in $\mathbb{R}^3$ consisting of all vectors with third entry 0 (the xy-plane).

(d) No. The vectors are linearly dependent because

$$v_1 + v_2 - v_3 = 0$$

is a nontrivial linear combination equal to zero.

Solution 11.

We require

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

This gives

$$c_1 + c_2 = 4, \qquad c_1 + c_3 = 3, \qquad c_2 + c_3 = 5.$$

Adding all three equations gives $2(c_1 + c_2 + c_3) = 12$, so $c_1 + c_2 + c_3 = 6$. Subtracting each equation in turn from this,

$$c_3 = 6 - 4 = 2, \qquad c_2 = 6 - 3 = 3, \qquad c_1 = 6 - 5 = 1.$$

Check: $v_1 + 3v_2 + 2v_3 = (1, 1, 0) + (3, 0, 3) + (0, 2, 2) = (4, 3, 5)$. Therefore

$$\boxed{b = v_1 + 3v_2 + 2v_3} \qquad \text{and} \qquad \boxed{b \in \operatorname{span}\{v_1, v_2, v_3\}}.$$

Solution 12.

Suppose

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0.$$

Read the coordinates from bottom to top. Only v_4 has a nonzero fourth entry, so $c_4 = 0$. Then only v_3 has a nonzero third entry, so $c_3 = 0$; similarly $c_2 = 0$, and finally $c_1 = 0$. Thus the vectors are **linearly independent**.

(a) **Yes.** Four linearly independent vectors in $\mathbb{R}^4$, which has dimension 4, necessarily span $\mathbb{R}^4$.

(b) **Yes.** They are linearly independent and span $\mathbb{R}^4$, so

$$\boxed{\text{they form a basis of } \mathbb{R}^4.}$$

Solution 13.

For three vectors in $\mathbb{R}^3$, linear independence and spanning go together: three independent vectors span $\mathbb{R}^3$, and three dependent vectors span at most a plane.

(a) The standard basis of $\mathbb{R}^3$ is linearly independent and spans $\mathbb{R}^3$.

$$\boxed{\text{Basis}}$$

(b)
$$\begin{pmatrix}1\\2\\1\end{pmatrix} = \begin{pmatrix}1\\1\\0\end{pmatrix} + \begin{pmatrix}0\\1\\1\end{pmatrix}.$$

Hence the vectors are dependent, and their span is only the plane spanned by $(1, 1, 0)$ and $(0, 1, 1)$, so they do not span $\mathbb{R}^3$.

$$\boxed{\text{Not a basis: fails both linear independence and spanning}}$$

(c)
$$\begin{pmatrix}2\\4\\6\end{pmatrix} = 2\begin{pmatrix}1\\2\\3\end{pmatrix}.$$

Hence the vectors are dependent, and their span is only the plane spanned by $(1, 2, 3)$ and $(1, 0, 1)$, so they cannot span $\mathbb{R}^3$.

$$\boxed{\text{Not a basis: fails both linear independence and spanning}}$$

Solution 14.

(a)
$$u^T v = 1(1) + 1(-1) + 0(0) = 0,$$
so $u \perp v$.

(b) Since $\|u\| = \|v\| = \sqrt{2}$,
$$q_1 = \frac{u}{\|u\|} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}, \qquad q_2 = \frac{v}{\|v\|} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}.$$

(c)
$$q_1^T q_1 = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad q_2^T q_2 = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad q_1^T q_2 = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$
so
$$\boxed{\{q_1, q_2\} \text{ is orthonormal.}}$$

(d) For an orthonormal set, coefficients are obtained by inner products:
$$c_1 = q_1^T x = \frac{5+1}{\sqrt{2}} = 3\sqrt{2}, \qquad c_2 = q_2^T x = \frac{5-1}{\sqrt{2}} = 2\sqrt{2}.$$
Thus
$$\boxed{x = 3\sqrt{2}\, q_1 + 2\sqrt{2}\, q_2}.$$
Check: $3\sqrt{2}\, q_1 + 2\sqrt{2}\, q_2 = (3, 3, 0) + (2, -2, 0) = (5, 1, 0) = x$.

Solution 15.

(a)
$$e = \hat{p} - p = \begin{pmatrix} 0.25 - 0.10 \\ 0.55 - 0.60 \\ 0.05 - 0.20 \\ 0.15 - 0.10 \end{pmatrix} = \begin{pmatrix} 0.15 \\ -0.05 \\ -0.15 \\ 0.05 \end{pmatrix}.$$

(b)
$$\|e\|_\infty = \max_i |e_i| = \max\{0.15,\, 0.05,\, 0.15,\, 0.05\} = \boxed{0.15}.$$

(c) The largest absolute deviation, 0.15, is attained by two labels:
$$|e_1| = |e_3| = 0.15.$$
So labels 1 and 3 are tied for the largest deviation: the student over-predicts label 1 and under-predicts label 3 by the same amount.

(d) The Chebyshev norm measures the **worst-case deviation** of any single output label, rather than the total deviation across labels, so it tells us how badly the student can disagree with the teacher on any one class.

Solution 16.

(a) **True.** A set of mutually orthogonal nonzero vectors is linearly independent: taking the inner product of $c_1u + c_2v + c_3w = 0$ with u gives $c_1\|u\|^2 = 0$, so $c_1 = 0$, and similarly $c_2 = c_3 = 0$.

(b) **False.** $\mathbb{R}^2$ has dimension 2, so it cannot contain three linearly independent vectors. For instance, $(1, 0)$, $(0, 1)$, $(1, 1)$ are dependent, since $(1, 1) = (1, 0) + (0, 1)$.

(c) **True.** Every orthonormal set is linearly independent (see (e)).

(d) **False.** A basis need not be orthogonal or normalized. For example,

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

is a basis of $\mathbb{R}^2$ but is not orthonormal.

(e) **True.** Orthonormal vectors are mutually orthogonal and nonzero (each has norm 1), hence linearly independent by (a).

(f) **False.** Four mutually orthogonal nonzero vectors would be four linearly independent vectors in $\mathbb{R}^3$, which is impossible since $\mathbb{R}^3$ has dimension 3.